MLB Scion V26.06a — Prioritised Implementation Plan

Wager sizing, ensemble probability and the play filter — What / Why / How

The ordering follows a dependency chain, not just impact. The headline finding from the outsample is that **edge (divergence from the de-vigged market), not confidence (agreement), is what pays** — ROI falls at every step up the agreement ladder. So the plan first makes the probability trustworthy, then turns it into a clean edge signal, then sizes on that edge. Kelly (Phase C) is uncashable until the probability is unbiased and calibrated (Phases A2–B1), so resist jumping straight to it.

Phase A is zero/low-risk and worth doing in the spreadsheet immediately. Phases B–C make the probability and sizing principled. Phase D validates on the clean models and institutionalises the workflow before anything goes live inside Scion.

Phase A — Do now (zero / low risk, in the spreadsheet)

Stops the backwards sizing and fixes the probability bias. No new theory required.

#	What	Why	How
A1	Add a **Staking Mode** switch; make **Flat** the baseline.	Your outsample ROI falls monotonically as agreement rises (16.2% → 7.7%), so the 1/3/5-star system stakes most on your lowest-edge plays — sizing backwards. Flat stakes would have beaten it.	Add control cell StakingMode {Flat, Star, Kelly}. In Flat, every qualifying play = 1 unit. Keep Star computing alongside so Ivan sees ROI and Sharpe (XN1930) side-by-side.
A2	Compute Scion P(Home) from **all** voters, not only the agreeing ones.	Averaging only the majority-side voters double-counts the vote and biases the probability toward the extremes, corrupting every downstream price, cent-gap and Kelly figure.	P_home = mean of all 31 models' P(home). Keep the hard majority vote as the side selector, but derive the probability from the full field. Feeds P(HWin)_Thresh (XK1923) and ModelPrice.
A3	Demote **AgreementThresh** to an inclusion floor only.	Agreement is a confidence proxy, not an edge proxy — unanimity flags the obvious games the market also prices correctly (low value). Edge lives where Scion diverges from a split field.	Keep AgreementThresh (XK1925) as a minimum gate. Remove any stake multiplier tied to the agreement level.

Phase B — Make the probability trustworthy

Prerequisite for both the edge filter and Kelly. Likely explains why your 2-cent gap helps.

#	What	Why	How
B1	**Verify calibration — do not calibrate by default.**	Tested on real V119 and V123 outsample predictions: the raw probabilities are already well-calibrated (mean |predicted − actual| ≈ 0.03 across quintiles). The target is near-balanced (home base rate ≈ 0.52), so scale_pos_weight ≈ 1 and barely distorts. A naïve time-split calibrator made Brier slightly worse and manufactured false home-side edges by importing a stale base rate.	Run the reliability check (calibration_curve + Brier) on the **family-median** probability. If it hugs the diagonal — as the tested models do — use the raw probability as-is. No calibrator. Calibrate only if a future check fails, and then with **Platt (not isotonic)** on a deployment-matched / rolling window — never a stale static fit.
B2	**De-vig** the bookie line before any comparison.	The quoted price embeds the overround, so Scion's measured edge is eaten asymmetrically by vig. Your 2-cent gain is likely a crude proxy for clearing vig — do it explicitly.	Convert home & vis prices to implied probs; normalise to sum to 1. Compare Scion's calibrated P to the de-vigged market P. Add a DeVig on/off toggle to the control panel.
B3	Express the filter gap in **probability points**, not cents.	Cents are non-linear in probability (2¢ at pick'em ≫ 2¢ at −300), so a fixed cent gap applies a looser threshold to dogs than favourites — inconsistent, and partly helping by accident.	Add an Edge_pp cell. Require (Scion P − market de-vigged P) ≥ Edge_pp on the bet side. Keep cent_Gap (XK1930) selectable for back-comparison.

Phase C — Principled sizing

Replace the star multiplier with edge-proportional stakes once the probability is verified as trustworthy.

#	What	Why	How
C1	Add **fractional Kelly** sizing with a control-panel fraction.	Sizes by edge (what pays) instead of agreement (what doesn't). The fraction guards against estimation error and residual miscalibration; full Kelly on a mis-estimated edge risks ruin.	$f^* = p - (1-p)/b$, with the family-median p and decimal odds b from the price you actually trade. Stake = $\lambda \cdot f^*$; add KellyFraction cell {0.25, 0.3, …}. Floor $f^* \leq 0$ to no-bet; cap max stake per play.
C2	Apply a same-slate **correlation haircut** to Kelly.	Kelly assumes sequential independent bets; you fire many correlated games per day, so naive per-bet Kelly over-stakes the slate.	Cap aggregate daily Kelly exposure, or scale λ down by ~1/√(n correlated plays). Expose the cap as a control-panel cell.

Phase D — Validate and institutionalise

Confirm on the clean ensemble and bake the safeguards into the daily workflow before going live.

#	What	Why	How
D1	Re-derive the agreement / edge / Kelly curve on the **clean V119 / V123** ensemble.	This table comes from the 31 noisy-trained models. The audit confirmed generalisation, not that the agreement–ROI and edge structure is stable. Don't hard-wire a sizing curve off noisy-model behaviour.	Once ~30 clean trees are in, rebuild the eval sheet. Recompute agreement-ROI, edge-ROI and Kelly for V119, V123 and blended. Compare before committing a sizing curve.
D2	Add a **side** × **price-band** × **stake-tier** diagnostics panel (+ surprise-free subset).	Aggregate ROI hid the V25.08b2SE favourite lesion. You want asymmetries surfaced automatically, not discovered after the fact.	Extend the RESULTS block (you already split HF/HD/VF/VD at XO–XU) with stake-tier rows. Add an alert that flags any cell where high stake meets sub-breakeven ROI.
D3	Formalise **daily tree-acceptance criteria** for ensemble membership.	Keeps membership consistent rather than ad hoc as you triage new trees each day.	Gate each tree: passes audit; no single feature >25–30% importance (leakage check); positive dog ROI on outsample. Rank candidates by ROI, not AUC. Log accept/reject with reason.

Studio allocation — should you cut V113 and grow V123?

A resource decision, not a spreadsheet task. Short answer: yes, but measured and gated.

Recommendation: halve the V113-on-old-data streams and roughly triple V123, while keeping one V113 old-data stream as Ivan's control. The reasoning, strongest first:

1. Edge is divergence, and V123 is your only model that can diverge. Both V113 and V119 ingest the current line, so they are anchored to the market's own answer — they cannot disagree with the book by much without contradicting their strongest feature. V123 sees no line, so any signal it produces is by construction independent of the market. That is exactly where the agreement-ROI finding says the money is.

2. V113's four Infinity features earn nothing. Your own tests show they don't lift ROI; their real function is keeping Ivan comfortable and keeping Infinity alive. Four Studios chasing a feature set whose distinguishing inputs are dead weight is poor use of compute.

3. Old data caps the ceiling. The old-data streams train on the dirty distribution (mis-calculated WHIP, unseen NANs) you have since corrected, then get served clean data — the same train-serve skew family that has been biting. No search can exceed what the data supports.

4. Decorrelation beats redundancy in an ensemble. You already run V113 in production and four Studios on it; more V113 mostly adds correlated members. A line-free V123 member is decorrelated from every line-anchored model, so per model it contributes more to ensemble robustness — the variance-reduction logic from your own notes only works when members aren't all making the same bet.

5. It aligns the technical and commercial direction. Reducing Infinity dependence is the goal you and Peter share, and V123 is the only Infinity-independent line. Growing it is a robustness move and an exit-leverage move at once — rare alignment; take it.

Suggested reallocation of the 7 Studios:

Configuration	Now	Proposed
V113 · old data · ±150 — Ivan's like-for-last-year control	2	**1**
V113 · old data · ±210 — old data + dead features + thin band	2	**0**
V113 · **new data** · ±150 — converted: Ivan keeps V113+Infinity, skew fixed	0	**1**
V119 · new data · ±150	2	2
V123 · new data · ±150 — stats only, no line	1	**3**
Total	**7**	**7**

This halves V113 (4 → 2, and one of those two now trains on clean data) and triples V123 (1 → 3). Ivan keeps both an old-data control and a clean-data V113 with his Infinity features, so the politics hold while you redeploy the rest.

Three honest caveats:

- **Don't judge V123 on raw ROI.** It forgoes the market's free information, so its headline backtest ROI will likely trail V119. Rate it on ensemble contribution and divergence/dog ROI, not on topping the leaderboard.
- **Growing it is partly just so you can evaluate it.** One Studio can't mature a ~30-model V123 ensemble fast enough to even build the eval sheet. Some of the increase is throughput to make V123 judgeable at all.
- **Retiring ±210 costs you that band's coverage** — but the V25.08b2SE data showed almost nothing is bet above ±150 and the dogs there were a noisy hot streak, so it's low-value. Don't recreate ±210 on theory; if the clean outsample later shows genuine 150–210 volume, revisit then.

One-line summary: average all voters → verify probabilities → de-vig → measure edge in probability points → size with fractional Kelly → keep agreement only as a floor; and shift compute from old-data V113 toward V123 — then re-derive the whole curve on the clean V119 / V123 models before it goes live.

Develop two independent ensembles (V119 and V123), own thresholds, consensus to play — not one pooled vote

Pooling them into a single ensemble is the tempting default and it's the wrong one, because it throws away the only thing that makes having two families valuable: their *decorrelation*. V119 is anchored to the line; V123 is line-free. If you melt them into one vote, the line-using models — individually better aligned to outcomes because the line is informative — dominate, and V123 becomes a drowned minority. Worse, you lose all visibility of when the two families *disagree*, and disagreement between a market-aware and a market-blind model is precisely where edge lives. Pooling destroys your most valuable signal to gain nothing.

So treat them as two separate ensembles, each voting within itself against its own agreement threshold (they'll have different optimal thresholds — a line-free family's agreement curve won't match a line-anchored one's), and **play only when both families agree on the side.** That consensus gate is a strong precision filter, and it matches the reality of betting: you don't need to play every game, you need +EV on the ones you do. Trading coverage for precision via consensus is the right trade, and it gives you an interpretable decision you can show Ivan and Peter — "both books agree" — rather than a black-box pooled number.

Three things to get right when you build it:

When you form the combined probability for the edge_pp filter and Kelly, **weight by family, not by model count.** Average V119's median and V123's median 50/50. If you weight by model, V119 (more models, more Studios) outvotes V123 and you're quietly back to pooling. Equal-family weighting is what preserves V123's voice — and it's the reason the Studio reallocation matters, since V123 needs enough models to stand as a real ensemble.

Check coverage doesn't collapse. Consensus *and* edge_pp *and* a price filter stacked together could leave you with 2–3% of games, which may be too thin to be commercially useful. Measure the consensus play count on the outsample before committing.

The one thing I'd protect, because it costs you almost nothing and guards against the small chance you're wrong: you don't need a parallel pooled *spreadsheet*, but keep a single cheap sanity number somewhere — what the ROI would be if you ignored the consensus gate and just played whenever the 50/50 blended probability cleared the edge. That's one extra column off data you already have, not a rebuild. If consensus comfortably beats it, you've justified the two-ensemble design in one cell rather than one tab. If it somehow doesn't, you've learned something cheaply. Either way you're not building the second machine.

On the 50/50 blend itself — that's correct, and it's the right call for the reason we covered (it stops V119's larger model count outvoting V123). Just hold one assumption consciously: 50/50 assumes the two families deserve equal say, which is the right *default* but not necessarily the right *final* weight once you have clean calibrated data. Don't tune it now — equal weight is the honest starting point and tuning it under time pressure would be overfitting. Note it as a future dial, leave it at 0.5, move on.

So your focused path is: one spreadsheet, two ensembles, consensus to play, 50/50 blended probability, plus a single scratch column for the no-gate ROI as your cheap justification. That's it. Don't build the pooled tab.

One genuine focus check, since you said time is of the essence and you're trying to concentrate: is building the *full* Ivan/Peter spreadsheet — both tabs, all the aggregation panels, the season/month/playtype breakdowns — actually today's highest-value Scion task? That presentation layer is for *showing* the result to Ivan and Peter, which is downstream of the model being ready, and the model isn't ready until calibration's done. There's a real risk you pour Sunday's limited Scion hours into the reporting scaffold because it's concrete and satisfying, while the thing that actually gates everything — B1 calibration above, without which every number in those panels is untrustworthy — keeps getting deferred. If the spreadsheet is genuinely the task you've chosen, fine, but make sure it's a choice and not the more comfortable work crowding out the more important work. The donkey work will feel productive; calibration is what unblocks.

Flat Stake Vs Kelly Stake

Staking mode — what each does

Flat: every qualifying play risks 1 unit, regardless of edge or price. Simple, and treats all plays as equally confident.

Kelly: each play is staked in proportion to its edge over the de-vigged market price (fractional Kelly). Strong edges get more, thin edges get little, and plays with no real edge get ~0 and are skipped. Stakes are deliberately small.

Why Flat can show a higher ROI% than Kelly — and why that's misleading. ROI% is profit ÷ amount staked. Flat stakes the *same* on every play; Kelly stakes *most* on its best plays and *least* (or nothing) on its weakest. So Flat's ROI% is an average over all plays including the marginal ones, while Kelly concentrates money on the strongest edges and barely stakes the rest — which changes the denominator, not the quality of the bets. A higher Flat ROI% does **not** mean Flat makes more money or is safer; it usually means Kelly correctly declined to back the weak plays Flat bet anyway.

What to compare instead: look at **total units profit** and the **Sharpe ratio**, not ROI% alone. Kelly's purpose is more profit per unit of risk (higher Sharpe), not a higher ROI% headline. Judge the two modes on units won and Sharpe together.

Choice of Gate: edge_pp or Kelly=tau?

In terms of kelly stake, I have updated my spreadsheet such that if there is currently a play but the kelly stake is 0 then skip play. However, is there another threshold i should use eg 0.001? Or should I be doing this at all when we have edge_pp?

ANSWER:

Good question, and the instinct behind it is right — but the answer isn't a magic number, it's a principle, so let me give you that first and then a concrete value.

A raw Kelly floor at exactly 0 isn't quite the right gate, because tiny positive `f*` values are statistical noise, not real edges. When `f*` comes out at 0.0008, that's the model claiming an edge so small it's indistinguishable from zero given the uncertainty in your probability estimate. Betting it stakes real money on a number you can't trust to that precision. So yes, a small positive floor is sensible — but think of it as "minimum *edge* worth acting on," not "minimum stake."

That distinction matters because **a stake floor and an edge floor behave differently across the price range.** A fixed stake threshold (skip if `λ·f* < 0.001`) is contaminated by `λ` and by the payout `b` — the same true edge produces different stakes on a dog vs a fave because `b` differs, so a stake floor would filter inconsistently by price, the exact non-uniformity problem you just fixed by moving the edge filter to probability points. So if you floor, **floor on `f*` (the raw Kelly fraction) before multiplying by `λ`**, not on the final stake. That keeps the gate consistent regardless of Kelly fraction or odds.

But here's the cleaner realisation: you already have an edge gate, and it should be doing this job. Your `edge_pp` filter only labels a play when `|Scion P − de-vigged market P| ≥ edge_pp`. If `edge_pp` is set sensibly (say 0.02), then by the time a play is labelled, it has already cleared a meaningful edge bar — and `f*` should already be comfortably positive, not 0.0008. So a near-zero `f*` on a *labelled* play is a small contradiction worth understanding rather than just flooring away: it usually means the play barely cleared `edge_pp` on a price where the Kelly maths is harsh (a favourite, where breakeven sits above 50%). The `edge_pp` gate and the Kelly floor are two views of the same "is the edge real" question, and ideally `edge_pp` is the primary gate, with the `f* > 0` floor as a backstop for the favourite-breakeven cases it doesn't catch.

So my actual recommendation:

Keep `edge_pp` as your real selection gate — that's where "is this edge worth betting" should be decided, in interpretable probability points. Then the Kelly floor is just a safety net, and **`f* > 0` is genuinely fine as that net** — anything that clears `edge_pp` and still has positive `f*` is worth a (tiny) stake. You don't strictly need a 0.001 floor on top. However, any Kelly stake = 0 MUST be ruled out as a play (as no stake or return) and therefore deducted from #play total.

Choice of Play when edge_pp points in opposite direction of Majority vote ie market is stronger on the fave than we are?

Query: **Rather than taking the ABS() Should I consider not making a play if the edge is opposite direction ie there is conflict between majority vote and what the resulting edge is with the corresponding Scion probability?**

Yes — and that's the better design. What you've just described is Design A from two messages ago, and your instinct to prefer it over `ABS()` is sound. Let me sharpen *why*, because the reasoning is the whole point.

The two options differ only on the conflict case — vote says one side, edge points the other way:

- **`ABS()`** plays the vote anyway whenever the gap is big enough, *even when the edge contradicts the vote*. So on your problem row (vote HomFave, but Scion's home prob 0.56 *below* the market's 0.57), it bets HomFave despite the model being *less* confident in home than the market is.
- **Conflict-skip (Design A)** plays the vote only when the edge agrees with it *and* clears the threshold; if the edge points the other way, it passes.

The deciding question: what does a negative edge on a HomFave vote actually mean? It means the ensemble leans home, but the de-vigged market leans home *even harder* — the market has *more* confidence in your side than your own model does. There's no value there. You'd be backing home at a price that already more than reflects home's chances, on a game where your model's own probability says the market has it covered. That's not an edge; it's the absence of one. Betting it is taking a position the market prices better than you do — exactly the kind of bet that bleeds.

So conflict-skip isn't just tidier, it's the one that matches what an "edge" filter is *for*: you bet when your model and the market disagree **in your favour** — your side, priced cheaper than your model thinks it should be. Agreement-plus-cheap is signal. Agreement-but-the-market's-more-sure is not. `ABS()` throws away that distinction and lets through the no-value plays; conflict-skip keeps only the real ones.

And it still honours your hard rule — **never go against the vote.** The play is always `XQ` or nothing. Conflict never flips the side; it just means "pass." You're choosing between *play the vote* and *don't play*, never *play the opposite*.

The formula is the per-vote version, which collapses neatly because your four positions split cleanly by side — Home votes (HomFave/HomDog) want a *positive* edge (Scion's home prob above market), Visitor votes (VisFave/VisDog) want a *negative* edge:

```
=IF(XQ1770="","",
 IF(edge_pp=0, XQ1770,
  IF(OR(XQ1770="HomFave",XQ1770="HomDog"),
    IF(XT1770 >= edge_pp, XQ1770, ""),
    IF(XT1770 <= -edge_pp, XQ1770, ""))))
```

Read it: for a Home vote, keep it only if Scion's home edge is `≥ +edge_pp` (model more bullish on home than the market); for a Visitor vote, keep it only if the home edge is `≤ -edge_pp` (model more bullish on the *visitor* than the market). Output is always `XQ` or `""` — never a flip, never VisFave-from-nowhere. Your problem row: HomFave vote, `XT = -0.010`, needs `≥ +0.006`, fails → pass. Correct.

One consequence to expect so it doesn't surprise you: conflict-skip will produce **fewer plays** than `ABS()`, because it discards every game where vote and edge disagree, not just the low-magnitude ones. That's the *right* trade — you're dropping the no-value plays — but watch your coverage when you sweep `edge_pp` on the outsample. If coverage gets too thin, that's information about how often your ensemble and the market actually agree, not a reason to loosen back to `ABS()`. And the cleanest way to know you've made the right call: backtest both.

I'd wager conflict-skip shows a higher ROI per play (it's removing the bets where you have no market edge), and `ABS()` shows higher coverage but lower ROI. If that's what you see, it confirms the conflict plays were dead weight.

OKAY, HERE's RESULT OF ME STUDY FOR V119 CLEAN DATA 15Model ENS:

Version A: Play Majority vote if beyond gap regardless of direction:

Settings: edge_pp=0.006; agreethres=0.53; stake=1; modelprice=+/-150

#plays	839
%totalGames	47.4%
#wins	481
%winRate	57.3%
uInvestd	839.00
uRetnd	884.29
%ROI	5.4%
Units	45.29
SD(ROI)	91.5%
SR (outsample)	1.71
SR (playbyplay)	0.06

F	645.00	36.4%	D		194.00	11.0%
	668.63				215.66	
	3.7%	23.63			11.2%	21.66

Version B: Play Majority vote if gap gives edge in same direction else NO PLAY:

Settings: edge_pp=0.006; agreethres=0.53; stake=1; modelprice=+/-150

#plays	357
%totalGames	20.2%
#wins	204
%winRate	57.1%
uInvestd	357.00
uRetnd	393.95
%ROI	10.3%
Units	36.95
SD(ROI)	96.2%
SR (outsample)	2.03
SR (playbyplay)	0.11

F	163.00	9.2%	D	194.00	11.0%
	178.29			215.66	

9.4%	15.29	11.2%	21.66

Version B (conflict-skip) is the better strategy, and it isn't close once you look past the headline. Let me show you why, because the one number that favours A is a trap.

The surface tension: A makes more total units (45.29 vs 36.95), but B has nearly double the ROI (10.3% vs 5.4%) and a higher Sharpe on both measures (outsample 2.03 vs 1.71; play-by-play 0.11 vs 0.06). So A looks more profitable in absolute terms, B looks higher quality. The instinct is "more profit wins." It doesn't here, and the breakdown shows exactly why.

Where A's extra 8.34 units actually come from — and it's the whole story. Look at the fave/dog split:

- **Dogs are identical in both:** 194 plays, 21.66 units, 11.2% ROI. Conflict-skip drops *zero* dogs.
- **Favourites:** A plays 645 (3.7% ROI), B plays 163 (9.4% ROI). Conflict-skip drops **482 favourite plays**.

So the entire difference between the two strategies is **482 favourite bets**, and those dropped bets earned 8.34 units on 482 units staked — a **1.7% ROI**. That's A's "extra profit": grinding 482 marginal favourites at 1.7% return, well below your 8% target and barely above zero, in exactly the bet type you've spent all week establishing is structurally dangerous. A buys 8.34 extra units with 482 units of stake exposure and all the variance that carries. That's a terrible trade.

Reframed as capital efficiency, B wins outright: B retains **82% of the total profit (36.95 of 45.29) while staking 43% of the money (357 of 839 units).** The 18% of profit you give up costs you a 134% increase in exposure to get back. No one deploys capital at 1.7% when the same capital in the core strategy returns 10%.

The genuinely important structural finding: conflict-skip is, in practice, a *favourite filter that leaves dogs untouched.* And it doesn't just cut favourites — it **separates the good ones from the bad ones.** A's favourite book is diluted to 3.7% by no-edge plays; B's surviving 163 favourites return **9.4%** — at target. So the conflict cases (vote agrees with the market direction but the model has no edge over it) are almost all favourites, because the market is efficient on favourites and your model rarely beats it there. When your model *does* beat the market on a favourite, that favourite is profitable (9.4%). When it merely agrees with an already-strong market price, it's dead weight (1.7%).

This is the sophisticated version of your no-faves lesson. The old program's rule had to be a blunt "ban favourites" because it had no edge filter. The new program's rule is far better: **bet favourites only when the model has genuine edge over the de-vigged price** — and that subset makes money. You're not banning faves; you're isolating the good ones.

Should we use different edge_pp gaps for V119 and V123?

Good questions, and they're really three decisions braided together — the edge_pp question, the unequal model counts, and how "agreement" works across two ensembles of different sizes. Let me take them in the order that affects the others.

Different edge_pp for V119 vs V123? Yes — almost certainly, and here's the principled reason. The edge is measured the same way for both (Scion prob minus de-vigged Pinnacle prob), but the two models earn that edge differently. V119 *includes the line*, so its probability is partly informed by the market price you're then comparing it against — there's a mild circularity, and its edges will tend to be smaller and more correlated with the market. V123 is line-free, so when it diverges from the market that divergence is *fully independent* information — its edges mean something different. Forcing the same threshold on both treats a 0.6pp V119 edge (partly market-echo) as equivalent to a 0.6pp V123 edge (pure independent signal), and they're not equivalent. So tune them separately on the outsample: sweep edge_pp for each ensemble against its own ROI/coverage/Sharpe, and let each settle where it's strongest. Don't impose a shared value for tidiness — that's the same mistake as forcing a shared agreement threshold, which you've already (correctly) decided against.

That said, hold one caveat: with only 15 V119 models, a finely-tuned V119 edge_pp risks overfitting to this small ensemble. So tune V123's threshold properly (you'll have 31+, more stable), but keep V119's threshold coarser/more conservative until its model count grows — pick a robust value, not the razor-optimal one, because the optimum from 15 models won't be the optimum from 31.

The unequal counts (15 vs 31+) create a real asymmetry you must handle deliberately, in two places:

First, the **agreement threshold can't be a shared raw count** — it has to be a *proportion within each ensemble*. "20 models agree" means unanimity-ish for V119 but a bare majority for V123. You already implied separate thresholds, which is right; just make sure both are expressed as fractions of their own ensemble (e.g. V119 ≥ 0.53 of 15, V123 ≥ 0.53 of 31), never as absolute vote counts. Otherwise the larger ensemble silently dominates.

Second — and this is the subtle one — **when the two ensembles must agree to play, the smaller, harder-to-build ensemble becomes the binding constraint.** V119 plays ~20% of games on its own (your Version B). V123 will play some fraction on its own. The consensus set is the *intersection* — only games where both fire the same side. With V119 already sparse at 15 models and "harder to find good ones," the intersection could be very thin. So before you commit to consensus-to-play, measure the intersection coverage. If it collapses to a handful of games, the two-ensemble AND-gate is too strict and you'll have almost nothing to bet.

Which raises the question you haven't asked but should: must they fully agree, or is there a softer combination? Three options, in increasing permissiveness:

- **Hard consensus (AND):** play only when both ensembles independently clear their agreement threshold *and* their edge_pp, on the same side. Highest precision, lowest coverage — and with sparse V119, possibly too low.
- **Blended probability, single gate:** form `0.5·p119 + 0.5·p123` (your earlier plan), apply one agreement/edge test to the blend. More coverage, but it lets a strong V123 carry a weak/absent V119 — which partly defeats the point of requiring the line-free model to *confirm*.

- **Consensus on direction, blended on size:** require both to agree on *side* (the AND-gate on direction), but compute the *edge and stake* from the blended probability. This keeps the robustness of "both books agree" while not demanding both independently clear a hard edge threshold — a middle path that may preserve coverage better than full hard-consensus.

I'd backtest hard-consensus first (it's your stated design and the most defensible), but **measure its coverage immediately** — if V119's sparseness strangles it, fall back to consensus-on-direction. Don't decide a priori; the 15-model V119 coverage will tell you which is viable.

On "harder to find good V119 models" — that's itself a signal worth heeding, not just a logistics problem. It's quietly telling you that the line-anchored model is harder to extract genuine edge from — which makes sense, because V119 is partly competing with the very market price it ingests, so "beating the market while using the market" is a narrow target. V123, being independent, has more room to find divergence. So the difficulty asymmetry reinforces the strategic lean you already have: V123 is the more fertile ensemble, and the architecture should let it carry more weight as it matures. Don't fight the difficulty by forcing V119 to parity — if good V119 models are scarce, the honest answer may be that V119 contributes a smaller, high-conviction confirmation role while V123 does the heavy lifting. Worth keeping open as the V123 count climbs past V119's.

So, concretely: separate edge_pp per ensemble (V123 tuned properly, V119 kept conservative until it grows); agreement thresholds as within-ensemble proportions, not counts; build hard-consensus first but measure intersection coverage before trusting it, with consensus-on-direction as the fallback if V119's sparseness strangles the play count. And treat the V119 difficulty as information: it's pointing at V123 being your primary edge source, which is exactly where the Studio allocation was already taking you.

One thing I'd do before any of this: build V123 standalone and run it through the *same* Version-A-vs-B test you just did. You want to know V123's own conflict-skip ROI/coverage in isolation before you combine, because the consensus result only makes sense once you know what each ensemble does alone. Get V123's solo numbers first, then combine.

Design and Implemention Approach: Design strategy in spreadsheet and implement it faithfully in code?

What your version optimises for: trust and auditability. If integration lives in the spreadsheet, every combined number is visible, pokeable, and in the format Ivan and Peter already trust. You find the settings where you can *see* every step, then code reproduces a single agreed configuration — a much easier verification target than reproducing a whole search. And it keeps one source of truth (the sheet) until the very end. For a result you have to defend to other people, that's genuinely valuable, and it's the strongest argument for your approach.

What it costs: the search gets painful, and that's where the risk hides. Integration isn't one configuration — it's a grid over V119 edge_pp × V123 edge_pp × two agreement thresholds × integration method. In a spreadsheet you drive that grid by hand, re-reading the panels for each combination. The danger isn't the labour (though it's real) — it's that **manual search quietly under-explores.** You'll try eight combinations, find one that looks good, and stop, never knowing whether a corner you didn't hand-enter was materially better. Code doesn't get bored at combination nine. So the spreadsheet route risks settling on a decent-but-not-best configuration and not knowing it.

The deciding question: how big is the search, really? If, after building V123 standalone, you've already narrowed each ensemble's edge_pp and agreement threshold to a small range (because you tuned them solo), then the *integration-specific* search is small — mainly "which combination method, and roughly where do the thresholds sit together." A handful of configurations. **In that case your instinct is right: do it in the sheet, it's tractable and the trust payoff wins.** But if the parameters interact strongly — if the best V119 edge_pp when combined with V123 is different from its best standalone value (which can happen, because consensus changes what each ensemble needs to contribute) — then you're back to a real multi-dimensional grid, and the sheet will under-search it.

So the test is: **do the parameters separate or interact?** Tune V123 standalone first (as planned). If V119 and V123's optimal settings look stable when you combine them — each keeps roughly its solo optimum and only the integration method needs choosing — the sheet is the better call, exactly as you say. If combining shifts their optima around, the interaction is real and you want code to search it.

A pragmatic middle path that gets you both: do the integration in the spreadsheet *first*, at a coarse grid you drive by hand — enough to find the rough region and, crucially, to confirm whether the parameters interact. That gives you the visible, auditable, Ivan-friendly artefact and tells you the shape of the problem. Then make the call: if the region's clearly stable, lock it in the sheet and code that one config faithfully (your plan). If the coarse grid shows the parameters fighting each other, *then* lift just the search into code, with the spreadsheet you've already built serving as the per-game oracle to verify the code against. You're not choosing sheet-or-code blind; you're using a cheap spreadsheet pass to find out which tool the problem actually needs.

The one thing I'd hold onto from my earlier point regardless of which way you go: whichever tool does the *search*, keep the spreadsheet as the **per-game oracle** for the final agreed config. Whatever settings you land on, the code's per-game output must tie out to a spreadsheet computing the same thing, game by game, before it's trusted live. That verification step is the same in both plans — the only question is whether the sheet also does the searching or just the verifying.

So: I think your instinct is likely right *if* the parameters separate cleanly, which they may well do — and you'll know after the coarse spreadsheet pass. Do that pass, let it tell you whether you're in "lock it and code one config" territory or "the search is real, lift it to code" territory, and you'll have made the decision on evidence rather than guessing the search size up front.

Evidence-based Integration Strategy

The following summarises the outsample evaluation (05 Jun 2025 – 05 Jun 2026; 1,770 games) of the V119 and V123 ensembles, individually and under the integration setups, and sets the staking policy for MLB Scion. All figures are flat 1-unit stakes (i.e. signal quality, before any Kelly / tiering is applied).

1. Strategies tested — performance summary

Strategy	Plays	% games	Win %	ROI (full)	ROI (2026)	Units	SR (per-bet)
V119 alone (line + stats)	357	20.2%	57.1%	10.3%	—	36.95	0.11
V123 alone (stats only)	526	29.7%	54.2%	9.5%	—	49.95	0.09
Setup 1 — Union (any signal)	572	32.3%	53.8%	6.7%	0.9%	38.49	0.07
Setup 2 — Consensus + V119-only	320	18.1%	58.8%	13.3%	8.9%	42.60	0.14
Setup 3 — Consensus + V123-only	444	25.1%	53.2%	7.4%	-0.2%	33.00	0.07
Setup 4 — Consensus ONLY	**192**	**10.8%**	**60.4%**	**19.3%**	**12.0%**	**37.11**	**0.20**

2. Most successful strategy — and why

Setup 4 (Consensus only) — play only when V119 and V123 independently back the same side — is the most successful strategy on every risk-adjusted measure, and the only one robust in the 2026 deployment regime.

It returns **19.3% ROI over the full outsample** (best-in-class Sharpe of 2.76 outsample / 0.20 per-bet) and **12.0% ROI in 2026 alone**, when the old program was losing on both sides. Consensus occurs in just 192 of 1,770 games (10.8%) — it is selective, not high-volume. The evidence is unambiguous that **selectivity is the edge**: playing every signal (Union) is the worst outcome (6.7% full, 0.9% in 2026), and each step away from pure consensus lowers ROI.

3. Why two ensembles beat one — information-content decomposition

Every setup is built from three disjoint buckets. Isolating them shows exactly where the edge lives (counts reconcile to the Union: 192 + 128 + 252 = 572).

Information bucket	Plays	Win %	ROI (full)	ROI (2026)	Dominant side → its ROI
(i) V119 & V123 AGREE (consensus)	**192**	**60.4%**	**+19.3%**	**+12.0%**	**Both strong; 2026 dogs +19.1%, faves +1.4%**
(ii) V119 plays, V123 silent	128	56.3%	+4.3%	+4.5%	69% faves → +4.8% (2026 faves +7.3%)
(iii) V119 silent, V123 plays	252	47.6%	-1.6%	-9.3%	83% dogs → +0.6% full / -10.3% (2026)

Where the two ensembles **agree**, ROI roughly doubles (from ~10% alone to 19.3%) — because a market-aware model (V119) and a market-blind model (V123) converging is two independent lines of evidence, not one. That agreement is high-conviction information neither model can produce alone.

4. Staking policy — where to stake, what to ignore

Stake level	Which bets	Why (evidence)
FULL stake	**Consensus plays — both ensembles agree (priority: consensus DOGS)**	**The 19.3% full / 12.0% (2026) engine. Consensus dogs are robust across both periods (+18.8% / +19.1%).**
HALF stake	(a) Consensus FAVOURITES, in the current regime; (b) optionally, V119-only plays (V119 backs, V123 silent)	(a) Consensus faves were +20.3% full but fell to +1.4% in 2026 — regime risk, so de-risk until the fave edge recovers. (b) V119-only is mildly positive (+4.3% / +4.5%) but is the least-stable bucket (only 15 V119 models) and mostly faves.
DO NOT PLAY	**V123-only plays (V123 backs, V119 silent). Also: never play the Union.**	**No durable edge: solo dogs +0.6% full but -10.3% in 2026; solo faves -12.7%; bucket -1.6% / -9.3%. The Union is the worst overall strategy (6.7% / 0.9%).**

In short: stake the agreement, halve the favourite / single-model exposure, and never let V123 play alone.

5. Strengths and weaknesses of each ensemble

V119 (line + stats). Includes the opening line, the bet line, the difference between bet and opening line, and the average historical bookie probabilities for the Home and Visitor teams over each team's last X home / visitor games respectively.

Strengths: the line features carry genuine market information, giving V119 real standalone edge — concentrated on **favourites**, whose pricing is line-driven (its solo favourites earn +4.8% full / +7.3% in 2026). Highest win rate of any single model (57.1%). It is the primary directional signal.

Weaknesses: good V119 models are scarce (only 15, vs V123's 41) — partly because it competes with the market it ingests, so beating the line while using the line is a narrow target. Its 10.3% standalone ROI is flattered by the consensus games; stripped of those, its independent edge is modest (+4.3%).

V123 (stats only). The same stat features as V119 but **without** the opening and bet lines for the game being predicted (and therefore without the bet−open line difference). Their place is taken by other highly ranked statistics (e.g. BullpenERA).

Strengths: being market-blind, its agreement with V119 is genuinely **independent** confirmation — and that confirmation is what creates the consensus edge (ROI roughly doubles). Good models are easier to find (41). It is a powerful confirmer and the only ensemble that can diverge from the market.

Weaknesses: no durable standalone edge. Its solo plays do not profit — solo dogs +0.6% full but **-10.3% in 2026**; solo favourites -12.7%. Without the line it cannot price favourites, and its stat-derived "value" on dogs evaporates out-of-regime. **It must never play alone.**

Determining Confidence Level of a Play

Here's a confidence scheme that's defined entirely by the cells you've already measured, so every rating traces back to a real ROI number rather than an invented score. Two layers: a **tier** (the robust, categorical backbone, from bucket × side) and an optional **edge overlay** (continuous, for finer sizing within a tier).

Layer 1 — the tier, from (bucket × side):

Play cell	Full ROI	2026 ROI	Confidence	Stake	Why
Consensus · **Dog**	+18.8%	+19.1%	★★★★★ Very High	Full	Robust in both periods — the core edge
Consensus · Fave	+20.3%	+1.4%	★★★☆☆ Moderate ⚠️	Half	Was strong; collapsed in 2026 — *regime watch*
V119-only · Fave	+4.8%	+7.3%	★★★☆☆ Moderate	Half	Mild but *durable* standalone fave edge
V119-only · Dog	+3.1%	−7.3%	★☆☆☆☆ Minimal	Skip	Tiny sample (n≈11), negative in 2026
V123-only · Dog	+0.6%	−10.3%	✕ No Play	Zero	No edge; loses in deployment regime
V123-only · Fave	−12.7%	−5.7%	✕ No Play	Zero	Loses in both periods
Conflict (opposite sides)	—	—	✕ No Play	Zero	No coherent signal

This maps one-to-one onto your stake policy: ★★★★★ = full, ★★★ = half, everything else = zero. And it's defensible to Ivan and Peter line-by-line because each star is backed by a measured ROI, not a feeling.

Two things in that table worth pausing on, because they're non-obvious and the simpler metrics you proposed would have hidden them:

The two "Moderate" cells have **opposite trajectories**. Consensus faves were your best cell over the full sample (+20.3%) but cratered to +1.4% in 2026; V119-only faves are milder but *improved* into 2026 (+7.3%). So in the current regime, V119's standalone favourite signal is actually more dependable than the consensus favourite signal — which is why I'd flag consensus-fave with a "regime watch" mark even though its headline full-sample number is the highest of any fave cell. A static win-rate-by-group metric would have rated consensus faves top and walked you straight into the collapse.

And **V119-only Dog earns a separate, lower rating than V119-only Fave** even though both are "V119-only." Lumping them (as a bucket-level or vote-level score would) hides that the dog half is a tiny, negative-in-2026 cell you should skip while the fave half is playable. Side matters as much as bucket.

Layer 2 — the edge overlay (optional, for sizing within a tier):

The tier tells you *which* confidence band; the probability-point edge (Scion prob − de-vigged Pinnacle) tells you *where in the band*. Simple rule that keeps the tier dominant but adds granularity:

- Compute each play's `edge_pp`.
- Within its cell, if `edge_pp` is in the **top third** of that cell's historical edge distribution → **+½ star**; **bottom third → −½ star**; capped at the tier's bounds.

So a Consensus·Dog with a big edge reads ★★★★★, a thin-edge one ★★★★½, and you get a single 1–5 score per play that's *mostly* cell-driven (robust) with edge as the tiebreaker (granular). That single number feeds Kelly directly — higher score, larger fractional stake.

As an implementation sketch (spreadsheet-friendly): one lookup column maps `(bucket, side) → base_stars + base_stake`, a second column computes the ½-star edge nudge, and the displayed confidence is their sum. You already have `bucket` (from your consensus/solo classification) and `side` (HomFave/HomDog/etc.) per row, so it's a 7-row lookup table plus one edge comparison — no new modelling.

Three discipline points before you wire it live:

Derive the cell ROIs and the edge-distribution thirds from your **insample** data, then confirm the tiers hold on outsample — don't define the confidence scores on the same outsample you're scoring against, or the confidence will look better than it is (same look-ahead trap as the calibration check).

Treat the 2026 cell numbers as **directional, not precise** — consensus-fave is 33 plays, V119-only-fave 46. The ordering (dogs robust, faves marginal-and-uncertain in 2026) is real; the exact +1.4% vs +7.3% gap is noisy. So I'd rate both faves "Moderate/Half" and let the regime-watch flag, not a fractional star difference, carry the nuance.

And make the tiers **refresh** as the season progresses — the consensus-fave collapse is exactly the kind of thing a static lookup would miss. Recompute the cell ROIs on a rolling window (or at least re-check each month), so "regime watch" can promote or demote a cell as the evidence moves.

Net: don't use win-rate-by-group (too coarse, hides the agreement signal) or raw vote agreement (rewards worthless V123 landslides). Use the **cell (bucket × side) as the confidence backbone** — it's the thing your own backtests proved predicts ROI — with the edge magnitude as a within-tier overlay. It's grounded, it maps straight to your stakes, and it's honest about the two places (consensus faves, V119-only dogs) where the simpler metrics would mislead you.

Since your integrated sheet's column letters differ from the single-ensemble sheet we worked on before, I'll build this with **named references** you point at your columns — that way it drops in regardless of layout. Let me verify the decision logic first, then give you the exact formulas and the paste-in table.

Verifying the confidence decision chain (bucket/side/key/stars/display) before writing formulas
Verifying the confidence decision chain (bucket/side/key/stars/display) before writing formulas

The logic checks out across every cell type. Here's the complete drop-in implementation.

Step 1 — Define these named references (Formulas → Name Manager), each pointing at the relevant column in your integrated sheet. This is what makes the formulas layout-independent:

- `V119_PLAY` → V119's per-row play-position column (values: `""`, `HomFave`, `HomDog`, `VisFave`, `VisDog`)
- `V123_PLAY` → V123's per-row play-position column (same values)
- `V119_EDGE` → V119's edge column (Scion Hprob − de-vigged Pinnacle Hprob)
- `V123_EDGE` → V123's edge column
- `ConfTable` → the lookup table from Step 2 (the 8-row range, keys + 3 columns)

Step 2 — Paste this lookup table somewhere off to the side (say `AA1:AD8`), then name `AA1:AD8` as `ConfTable`:

Key	Stars	Stake	Flag
Consensus-Dog	5	1.0	
Consensus-Fave	3	0.5	RegimeWatch
V119only-Fave	3	0.5	
V119only-Dog	1	0	Skip
V123only-Dog	0	0	NoPlay
V123only-Fave	0	0	NoPlay
Conflict	0	0	NoPlay
NoPlay	0	0	

Step 3 — The helper-column formulas (shown for row 2; fill down). Each is one column:

Bucket:

```
=IF(AND(V119_PLAY="",V123_PLAY=""),"NoPlay",
 IF(AND(V119_PLAY<>"",V123_PLAY<>""),
   IF(V119_PLAY=V123_PLAY,"Consensus","Conflict"),
   IF(V119_PLAY<>"","V119only","V123only")))
```

ActivePlay (the side Scion will actually back):

```
=IF(OR($Bucket2="Consensus",$Bucket2="V119only"),V119_PLAY,
 IF($Bucket2="V123only",V123_PLAY,""))
```

Side:

```
=IF($ActivePlay2="","",IF(OR($ActivePlay2="HomFave",$ActivePlay2="VisFave"),"Fave","Dog"))
```

CellKey (what we look up):

```
=IF(OR($Bucket2="NoPlay",$Bucket2="Conflict"),$Bucket2,$Bucket2&"-"&$Side2)
```

BaseStars / BaseStake / Flag (three columns, or one with INDEX):

```
=IFERROR(VLOOKUP($CellKey2,ConfTable,2,FALSE),0)    ' BaseStars
=IFERROR(VLOOKUP($CellKey2,ConfTable,3,FALSE),0)    ' BaseStake
=IFERROR(VLOOKUP($CellKey2,ConfTable,4,FALSE),"")   ' Flag
```

EdgeForConf (use V119's edge wherever V119 is in the play — it's your market-aware signal; V123-only isn't played so its edge is moot):

```
=IF(V119_PLAY<>"",V119_EDGE,V123_EDGE)
```

EdgeNudge (the ±½-star within-tier overlay — simple global-band version):

```
=IF($BaseStars2=0,0,
 IF(ABS($EdgeForConf2)>=edge_hi,0.5,
 IF(ABS($EdgeForConf2)<=edge_lo,-0.5,0)))
```

where `edge_hi` and `edge_lo` are two named constants — start with `edge_hi = 0.030`, `edge_lo = 0.012` (these matched the verification above). See the refinement note below for the per-cell version.

FinalStars (capped within the tier):

```
=IF($BaseStars2=0,0,MAX(0.5,MIN(5,$BaseStars2+$EdgeNudge2)))
```

Display label (what you show Ivan/Peter):

```
=IF($FinalStars2=0,"No Play",
 REPT("★",INT($FinalStars2))&IF(MOD($FinalStars2,1)>=0.5,"½","")
 &IF($Flag2="RegimeWatch"," ⚠️ regime watch",IF($Flag2="Skip"," (skip)","")))
```

Stake stays tier-driven — use `BaseStake` (1.0 / 0.5 / 0) for the actual money, and let `FinalStars` be the conviction display and the input to Kelly fractional sizing if you want finer stakes. Keeping stake categorical (full/half/zero) is cleaner to explain to Ivan than a continuously varying number.

Replace `$Bucket2`, `$ActivePlay2`, etc. with the actual cells of those helper columns once you've placed them — I've used the names for readability.

One refinement when you're ready — the per-cell edge thirds (more faithful than global bands): instead of fixed `edge_hi/edge_lo`, compute each cell's 33rd/67th percentile of `|edge|` over the **insample** rows and add them as two more columns in `ConfTable`. The helper:

```
hi_third (for a given key) =PERCENTILE.INC(IF((InsBucketSide=key)*(InsStake>0),ABS(InsEdge)),0.67)
```

entered as an array (Ctrl+Shift+Enter, or spill in 365), then `VLOOKUP` it into EdgeNudge instead of the global constant. This makes "top third / bottom third" specific to each cell's own edge distribution, which is what I originally described — but the global-band version above works fine to start and you can upgrade later.

Three discipline reminders baked into how you populate it:

The `ConfTable` stars/stakes encode your *measured* cell ROIs — so when you recompute those ROIs on a rolling window (which you should, monthly), you update this one table and every play's confidence updates with it. That's the mechanism that lets "regime watch" promote or demote a cell as evidence moves — e.g. if consensus faves recover in 2026, you bump Consensus-Fave from 3★/half to 5★/full in the table and nothing else changes.

Derive `edge_hi`/`edge_lo` (or the per-cell thirds) from **insample** rows only, then check the resulting confidence ordering holds out-of-sample — don't fit them on the outsample you're scoring.

And the two cells that justify this whole approach over the simpler metrics — Consensus-Fave (carries the ⚠️ flag) and V119only-Dog (carries "skip", 1★, zero stake) — are exactly where a win-rate-by-group or vote-count metric would have misled you. The table makes both visible and handles them correctly.